

WASP-36b

R-2447

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_130115 · created 2026-08-22T23:42:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456307.7742 +/- 0.00042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1463 +/- 0.0094
Transit depth (Rp/Rs)^2	2.14 +/- 0.27 [%]
Orbital Inclination (inc)	82.58 +/- 0.44
Airmass coefficient 1 (a1)	0.999 +/- 0.01
Airmass coefficient 2 (a2)	0.0011 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#2 - [91, 175]
Optimal Method	PSF photometry
Transit Duration (day)	0.0728 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1463 ± 0.0094 vs 0.1368 (deviation 0.94σ)
Duration fit vs published	0.0728 d vs 0.0758 d (deviation 0.85σ)
Mid-time O–C	1.1 min
Depth SNR	14.4
Residual scatter	1.01 %

Light curve

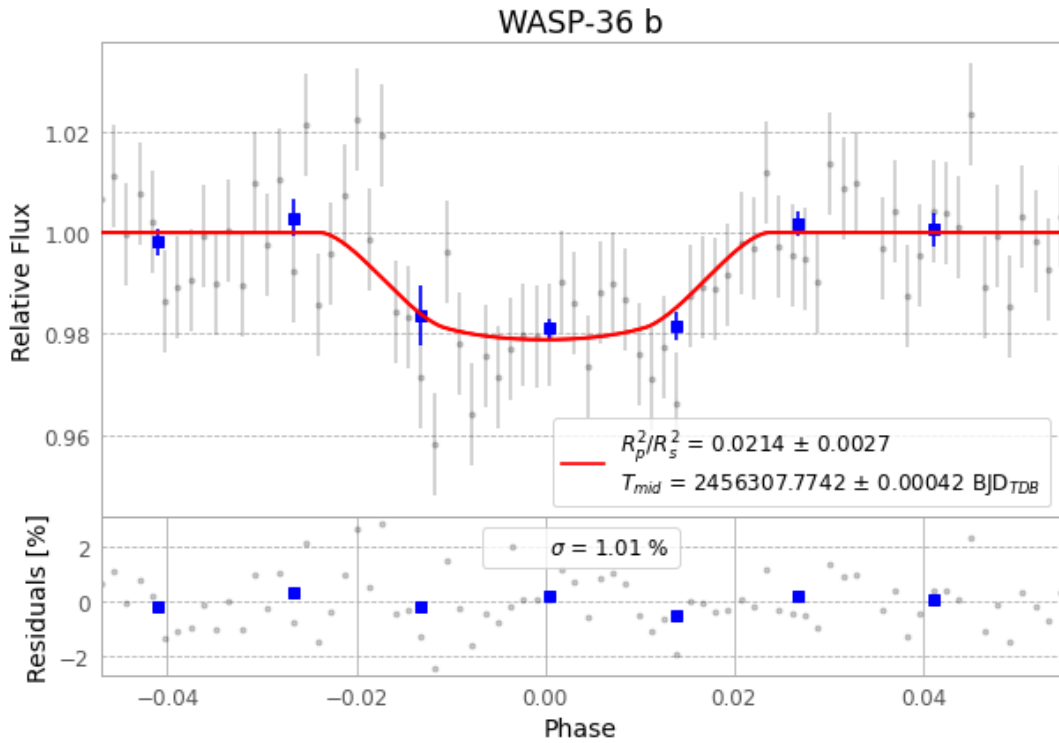


Figure 1 — FinalLightCurve_WASP-36 b_2013-01-14.png (SHA-256 7f1929addb0e5368...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [175, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8990bbd81ece443b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), WASP-36130115044212.FITS → WASP-36130115082713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130115.log (41597f52da2c...), FinalLightCurve_WASP-36 b_2013-01-14.png (7f1929addb0e...), FinalLightCurve_WASP-36 b_2013-01-14.csv (22a2415bc51c...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:42Z.